15. Structures between two adjacent cells which is an effective transport pathway- (2010 AIPMT)
**a. Plasmodesmata**
b. Middle lamella
c. Secondary wall layer
d. Primary wall layer

16. In active transport carrier proteins are used, which use energy in the form of ATP to
a. transport molecules against concentration gradient of cell wall
b. transport molecules along concentration gradient of cell membrane
**c. transport molecules against concentration gradient of cell membrane**
d. transport molecules along concentration gradient of cell wall

17. The main organelle involved in modification and routing of newly synthesised protein to their destinations is (AIPMT 2005)
a. Mitochondria b. Glyoxysomes
c. Spherosomes **d. Endoplasmic reticulum**

18. Algae have cell wall made up of (AIPMT 2010)
**a. Cellulose, galactans and mannans**
b. Cellulose, chitin and glucan
c. Cellulose, Mannan and peptidoglycan
d. Muramic acid and galactans

## UNIT IV – Plant Anatomy

1. The balloon – shaped structures called tyloses (NEET II – 2016 )
a. originate in the lumen of vessels
b. characterise the sap wood
**c. are extensions of xylem parenchyma cells into vessels**
d. are linked to the ascent of sap through xylem vessels

2. Cortex is the region found between (NEET II – 2016)
**a. epidermis and stele**
b. pericycle and endodermis
c. endodermis and pith
d. endodermis and vascular bundle

3. Read I – IV and find the correct order of components from outer side to inner side in a woody dicot stem (CBSE -AIPMT – 2015)
(I) secondary Cortex (II) wood
(III) secondary phloem (IV) phellem
a. III, IV, II and I b. I, II, IV and III
**c. IV, I, III and II** d. IV, III, I and II

4. You are given a fairly old piece of a dicot stem and a dicot root. Which of the following anatomical structures will you use to distinguish between the two? (CBSE -AIPMT 2014)
a. secondary xylem b. secondary phloem
**c. protoxylem** d. cortical cells

5. Heart wood differs from sapwood in (CBSE -AIPMT 2010)
a. the presence of rays and fibres
b. the absence of vessels and parenchyma
**c. having dead and non-conducting elements**
d. being susceptible to hosts and pathogens

6. The annular and spirally thickened conducting elements generally develop in the protoxylem when the root or stem is (CBSE -AIPMT 2009)
a. maturing b. elongating
**c. widening** d. differentiating

7. Anatomically fairly old dicotyledonous root is distinguished from the dicotyledonous stem by the (CBSE- AIPMT 2009)
a. absence of secondary xylem
b. absence of secondary phloem
c. presence of cortex
**d. position of protoxylem**

8. In barley stem, vascular bundles are (CBSE -AIPMT 2009)
a. open and scattered
**b. closed and scattered**
c. open and in a ring
d. closed and radial

9. Palisade parenchyma is absent in the leaves of (CBSE- AIPMT 2009)
**a. sorghum** b. mustard
c. soyabean d. gram

10. Sugarcane plant has (AIIMS 2009)
a. reticulate venation
b. capsular fruits
c. pentamerous flowers
**d. dump-bell shaped guard cells**

11. Vascular tissues in flowering plants develop from (CBSE- AIPMT 2008 & JIPMER 2012)
a. phellogen **b. plerome**
c. periblem d. dermatogen

12. The length of different internodes in a culm of sugarcane is variable because of (CBSE -AIPMT 2008)
a. short apical meristem
b. position of axillary buds
c. size of leaf lamina at the node below each internode
**d. intercalary meristems**

13. Passage cells are thin-walled cells found in (CBSE -AIPMT 2007)
**a. endodermis of roots facilitating rapid transport of water from cortex to pericycle**
b. phloem elements that serve as entry points for substances for transport to other plant parts
c. testa of seeds to enable emergence of growing embryonic axis during seed germination
d. central region of style through which the pollen tube grows towards the ovary

14. Which one of the following is not a lateral meristem (CBSE -AIPMT 2010)
a. interfascicular cambium



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